MATHEMATICS DEPARTMENT
FAITH · RIGHTEOUSNESS · WISDOM

cognia SCHOOL OF DISTINCTION 2024

مدارس دار الفكر
DAR ALFIKR SCHOOLS

GRADE 10 · ALGEBRA II · TOPIC 2

Lesson 2-6

The Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Journey of 6 stations, 60 minutes

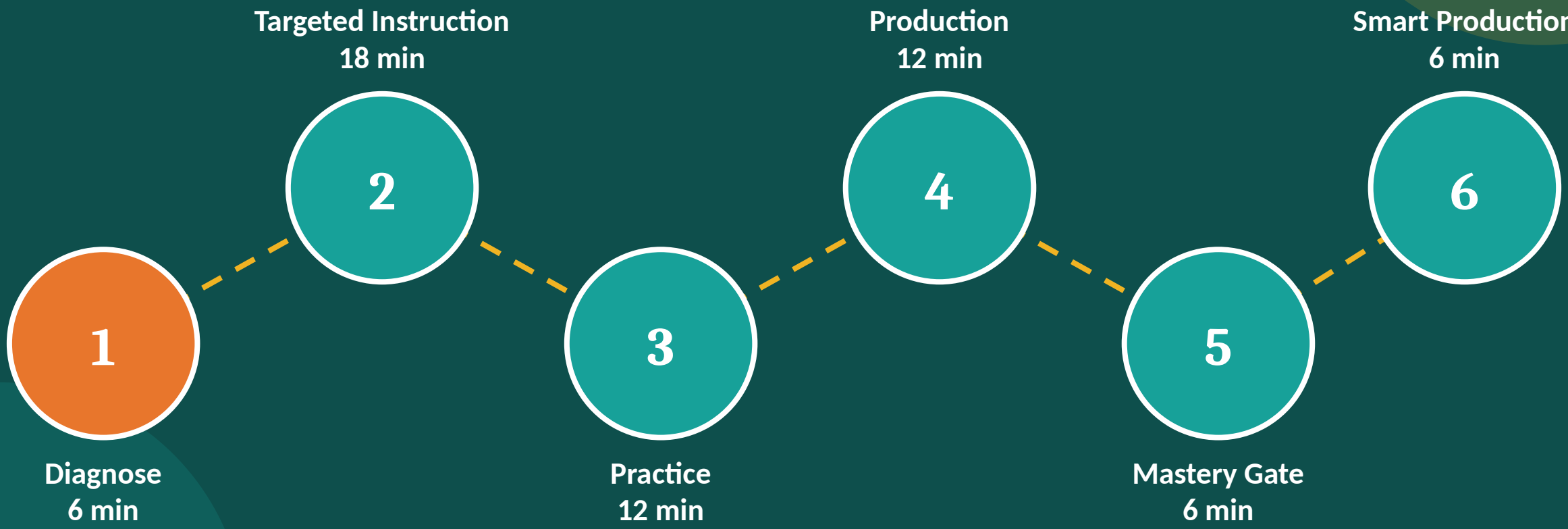
Quadratic Functions and Equations

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Today's journey

Six stations. Each one zooms in as we arrive.



What we are learning

OBJECTIVES

1. Analyze the discriminant to identify the number and type of solutions for a quadratic equation.
2. Solve quadratic equations using the quadratic formula, including those with nonreal solutions.

VOCABULARY

Discriminant **quadratic formula**

STANDARDS

HSN.CN.C.7 Solve quadratic equations with real coefficients that have complex solutions.

HSA.REI.B.4.A Use the method of completing the square to transform any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions. Derive the quadratic formula from this form.

HSA.REI.B.4.B Solve quadratic equations by inspection (e.g., for $x^2 = 49$), taking square roots, completing the square, the quadratic formula and factoring, as appropriate to the initial form of the equation. Recognize when the quadratic formula gives complex solutions and write them as a ± bi for real numbers a and b.

STATION 1 · DIAGNOSE · 6 MIN

Start here: three quick checks

1

Evaluate $b^2 - 4ac$ for $a = 2,\ b = -3,\ c = -5$

2

Simplify: $\dfrac{4 + \sqrt{20}}{2}$

3

Write $\sqrt{-36}$ using i

Done when

All three answers are written without a calculator, and you can say what goes wrong if the minus sign in Question 1 is lost.

STATION 2 · TARGETED INSTRUCTION · 18 MIN

The quadratic formula

$$ax^2 + bx + c = 0 \;\Rightarrow\; x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Write a, b, c with their signs first

EXAMPLE A

Solve: $x^2 - 5x + 6 = 0$

$a = 1,\ b = -5,\ c = 6$

$$x = \frac{5 \pm \sqrt{25 - 24}}{2} = \frac{5 \pm 1}{2}$$

$x = 3$ or $x = 2$

EXAMPLE B

Solve: $2x^2 + 3x - 5 = 0$

$a = 2,\ b = 3,\ c = -5$

$$x = \frac{-3 \pm \sqrt{9 + 40}}{4} = \frac{-3 \pm 7}{4}$$

$x = 1$ or $x = -\frac{5}{2}$

EXAMPLE C

Solve: $x^2 - 4x - 1 = 0$

$a = 1,\ b = -4,\ c = -1$

$$x = \frac{4 \pm \sqrt{16 + 4}}{2} = \frac{4 \pm 2\sqrt{5}}{2}$$

$x = 2 \pm \sqrt{5}$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

The discriminant

$\Delta = b^2 - 4ac$ $\Delta > 0$: two real $\Delta = 0$: one real $\Delta < 0$: two nonreal

EXAMPLE A

Classify: $x^2 + 6x + 9 = 0$

$\Delta = 36 - 36 = 0$

one real solution, $x = -3$

EXAMPLE B

Classify: $x^2 + x - 3 = 0$

$\Delta = 1 + 12 = 13 > 0$

two real, irrational solutions

EXAMPLE C

Classify: $2x^2 - 3x + 5 = 0$

$\Delta = 9 - 40 = -31 < 0$

two nonreal solutions

STATION 2 · TARGETED INSTRUCTION · 18 MIN

Nonreal solutions

$$\Delta < 0: \ \sqrt{-n} = \sqrt{n}\, i \qquad x = a \pm bi \ \text{(a conjugate pair)}$$

EXAMPLE A

Solve: $x^2 - 4x + 13 = 0$

$$x = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm 6i}{2}$$

$$x = 2 \pm 3i$$

EXAMPLE B

Solve: $2x^2 + 2x + 5 = 0$

$$x = \frac{-2 \pm \sqrt{4 - 40}}{4} = \frac{-2 \pm 6i}{4}$$

$$x = -\frac{1}{2} \pm \frac{3}{2}i$$

EXAMPLE C

Solve: $x^2 + 2x + 2 = 0$

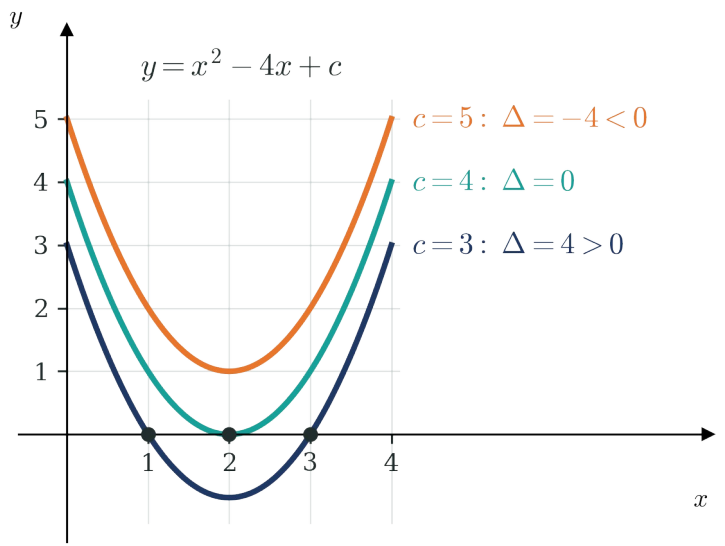
$$x = \frac{-2 \pm \sqrt{4 - 8}}{2} = \frac{-2 \pm 2i}{2}$$

$$x = -1 \pm i$$

STATION 2 · TARGETED INSTRUCTION · 18 MIN

The discriminant with an unknown

one real solution: $\Delta = 0$ two real: $\Delta > 0$ none: $\Delta < 0$



EXAMPLE A

One solution: $x^2 - 4x + k = 0$

$$\Delta = 16 - 4k = 0$$

$$k = 4$$

EXAMPLE B

One solution: $x^2 + kx + 16 = 0$

$$\Delta = k^2 - 64 = 0$$

$$k = 8 \text{ or } k = -8$$

STATION 3 · PRACTICE · 12 MIN

Give it a go: Practice, Apply, Investigate

PRACTICE

Secure the method

Solve: $2x^2 - 5x - 3 = 0$

Classify: $x^2 - 6x + 10 = 0$

Done when: a, b and c are written with their signs, the discriminant is found first, and each real solution is checked by substitution.

APPLY

Use it in context

Solve: $3x^2 - 6x - 2 = 0$

Solve: $x^2 + 6x + 25 = 0$

Done when: the answers are exact, the radical is simplified, every term of the numerator is divided by the denominator, and nonreal solutions are written in the form a ± bi.

INVESTIGATE

Find out why

For which k does $x^2 - 6x + k = 0$ have two real solutions?

Why does $x^2 - 5x + 6 = 0$ factor but $x^2 - 5x + 3 = 0$ not?

Done when: your argument names the discriminant, says what its value tells you about the solutions, and reaches an inequality for k.

STATION 4 · PRODUCTION · 12 MIN

Production: Jeddah Season ticket price

THE MODEL

Organisers of a Jeddah Season concert, part of the Vision 2030 entertainment programme, model the ticket revenue in riyals as a quadratic function of the ticket price in riyals, given below. These are model values, not real figures.

$$R(p) = -10p^2 + 400p \quad \text{(SAR)}$$

$$\text{Target revenue: } R(p) = T$$

1. Find the ticket prices that give a revenue of 2 000 SAR, exactly and to the nearest 0.01 SAR.
2. Show that a revenue of 4 000 SAR is possible at exactly one price, and state it.
3. A sponsor asks for 5 000 SAR. Use the discriminant to explain why no price works.

DONE WHEN

- Each task rewrites the revenue equation as a quadratic equal to zero and writes a, b and c with their signs.
- The discriminant is calculated and its sign gives the number of prices.
- Task 1 gives two prices, exact and decimal, and one is checked by substitution.
- Task 3 ends with a sentence in context and says the numbers are model values.

Hint for solving: divide every term by 10 before you use the formula.

STATION 5 · MASTERY GATE · 6 MIN

Mastery Gate: three questions to pass

1

How many real solutions does $3x^2 - 4x + 5 = 0$ have?

2

Solve: $x^2 + 6x + 4 = 0$

3

Solve: $x^2 + 2x + 17 = 0$

Pass mark

3 of 3 correct unlocks Station 6. Fewer than that: return to the slide for the question you missed, redo it, then re-take the gate.

STATION 6 · SMART PRODUCTION · 6 MIN

Smart Production: make one, then check it

CREATE

1. Write three quadratic equations of your own: one with a positive discriminant, one with a zero discriminant and one with a negative discriminant.
2. Swap with a partner. Find the discriminant of each equation and solve it with the quadratic formula.
3. Check one real solution by substitution, and say how the discriminant showed the type of solutions.

Done when: all three discriminants are correct, the solutions match their signs, and the check is written.

TIME TO CHECK

Exit ticket: write your answer on a sticky note and place it on the door.

Solve using the quadratic formula:

$$x^2 - 6x + 13 = 0$$

Next lesson: 2-7 Solving Linear-Quadratic Systems.

Exam connections

SAT

$x^2 - 8x + c = 0$ has exactly one real solution. What is c?

STEPS

1 $\Delta = (-8)^2 - 4(1)(c) = 64 - 4c$

2 $64 - 4c = 0$

3 $c = 16$

GAT

Which quantity is greater?

A: real solutions of $x^2 + 4x + 7 = 0$ B: 1

STEPS

1 $\Delta = 16 - 28 = -12 < 0$

2 A $= 0$ real solutions

3 $0 < 1 \Rightarrow$ B is greater

SAAT

The solutions of $2x^2 - 3x - 1 = 0$ are

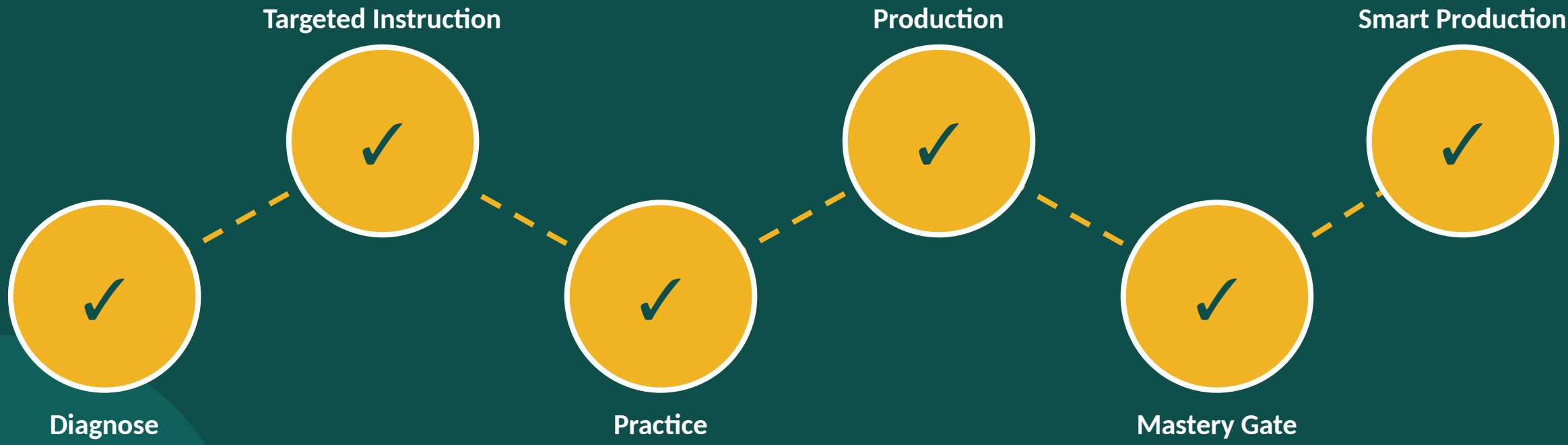
STEPS

1 $\Delta = 9 + 8 = 17$

2 $x = \dfrac{3 \pm \sqrt{17}}{4}$

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Journey complete



The discriminant $b^2 - 4ac$ tells you how many solutions and of what type, and the quadratic formula then solves any quadratic, giving a ± bi when the discriminant is negative.